Gentle Steps –
A Camera free home activity monitor for elderly care

Non-invasive 24GHz mmWave radar activity tracking and automated behavioral anomaly detection for safe independent living.

Zero Cameras or Microphones

LD2410c FMCW Radar

PROBLEM STATEMENT

By 2030, India will be home to nearly **194 million seniors** (about 13% of the population). More than ever, our elders are choosing the path of **independence**, preferring to live on their own terms rather than moving in with family members or kids

Some will move to retirement homes, but others prefer to stay in their own villas or apartments



PROBLEM STATEMENT

Balancing Safety with Personal Dignity

Privacy Intrusiveness

Traditional optical cameras cause discomfort and violate elderly privacy in bathrooms and bedrooms.

Undetected Emergencies

Falls or medical incidents inside private rooms often go unnoticed for hours without active monitoring.



THE NEED

The reassurance I need is simple, are they active, where are they and I getting the current information

ACTIVE

Evidence of normal movement during the day

WHERE

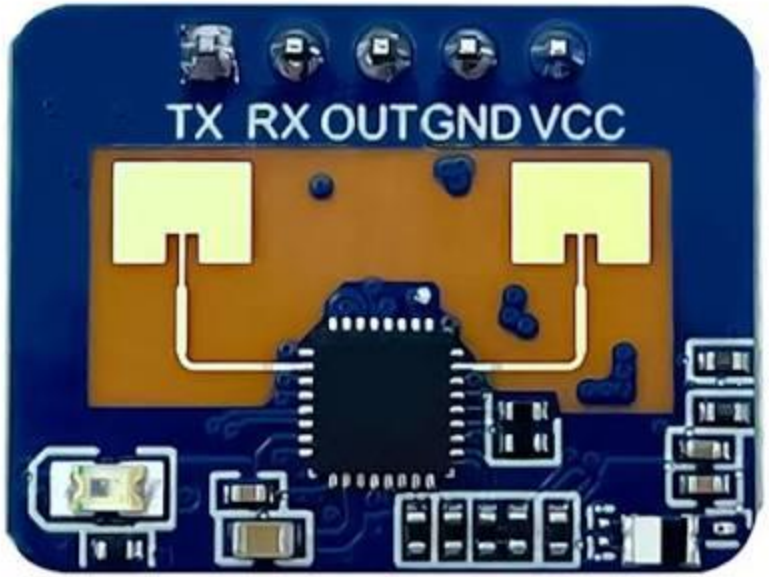
The room where they are located now

CURRENT

A visible last-received time stamp for ever update

THE SOLUTION

LD2410c Presence Sensor



Non-invasive 24GHz mmWave radar activity tracking and automated

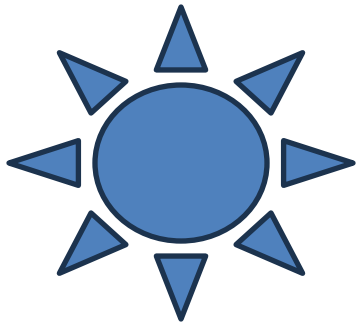
There are simillar sensor operating at 60GHZ

Is 24GHz radar based sensor Safe to be used indoor ?

Ans: It is absolutely safe as 24GHz is far below the ionizing frequency

Energy per Quantam (photon) = Planks Const X Frequency
= 6.62607015 × 10-34 X 24 X 10 to the power of 9
Energy = 1.6 X 10 to the power of –23 Joule

Energy per Quantam (photon) of Visible light = h x 484 X 10 (+12)
6.62607015 × 10-34 X 24 X 10 to the power of 12
3.21 X 10 to the power of –19 Joules

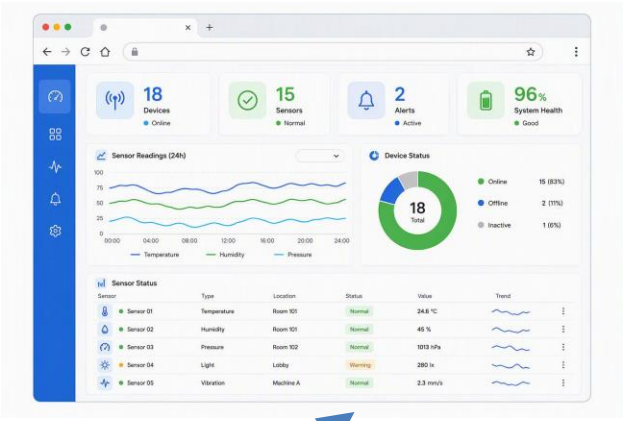


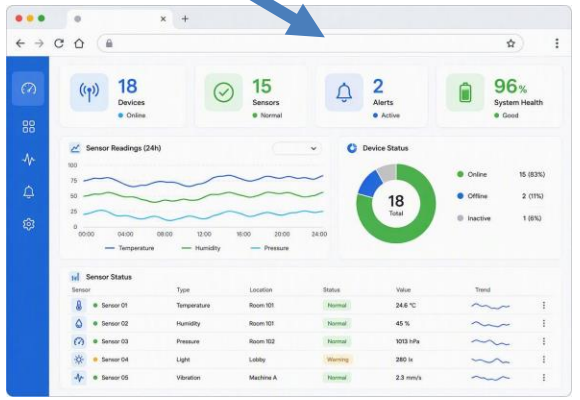
Cluster of sensor



Processing engine

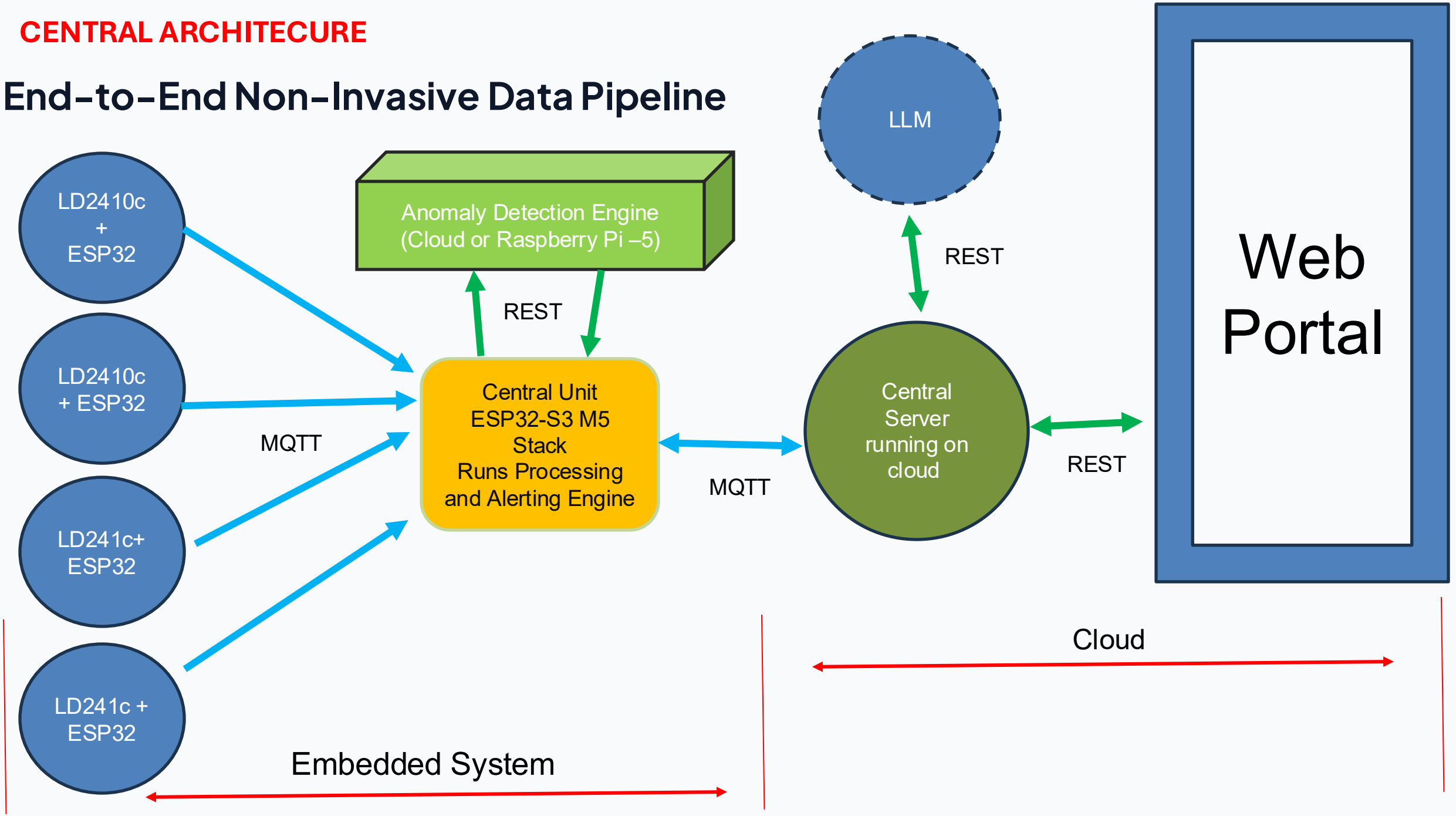
Web Server in Cloud

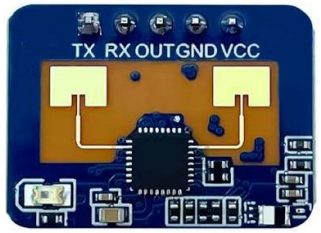




CENTRAL ARCHITECURE

End-to-End Non-Invasive Data Pipeline





LD2410c



WiFI

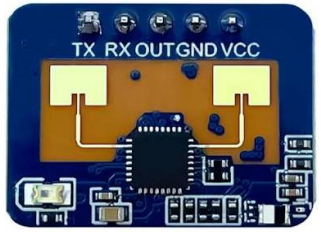
Cloud (AWS)

WiFi



WiFi

LD2410c

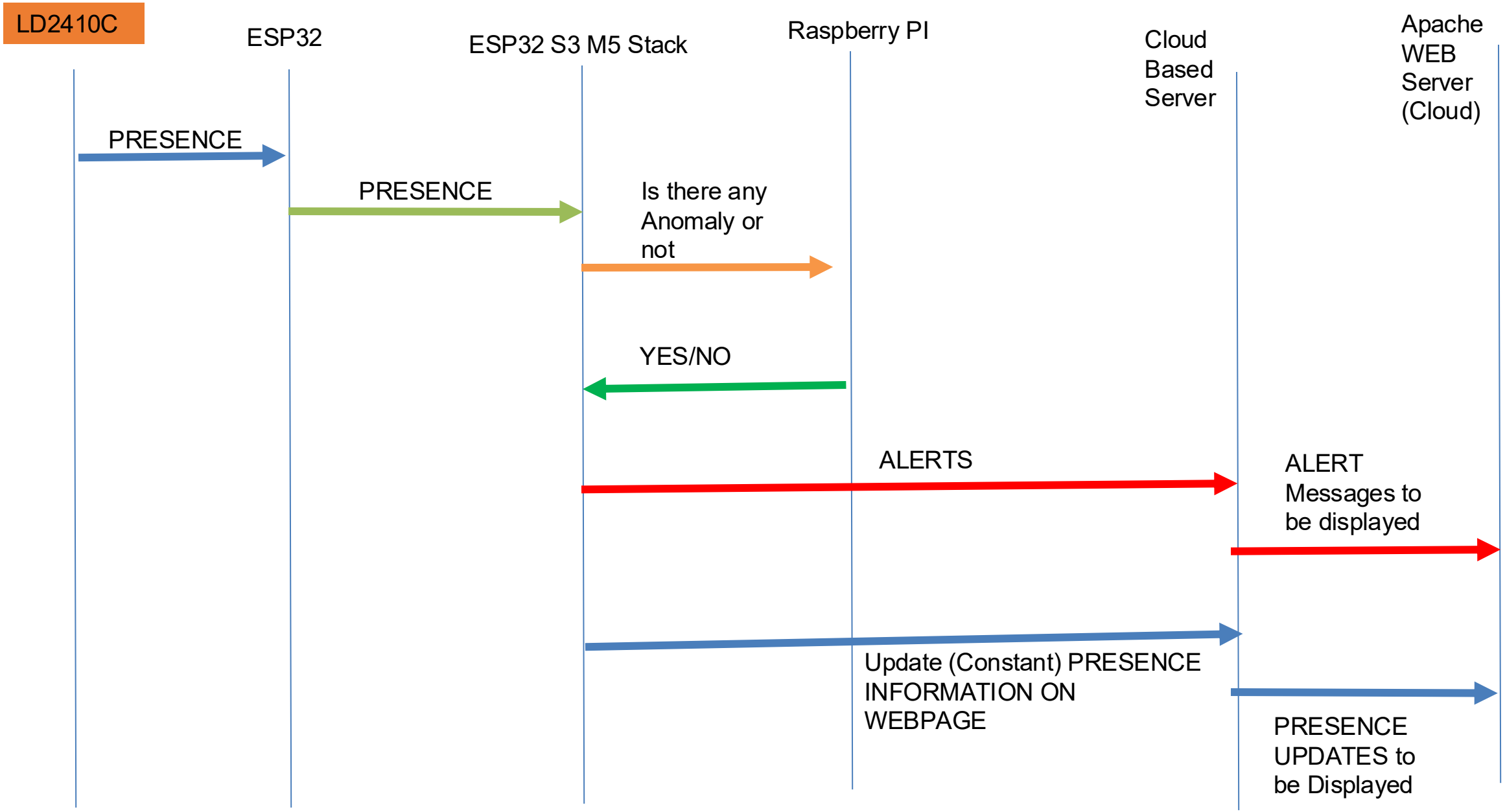




WiFI



Raspberry PI


LD2410C
ESP32
ESP32 S3 M5 Stack
Raspberry PI
Cloud Based Server
Apache WEB Server (Cloud)
PRESENCE
PRESENCE
Is there any Anomaly or not
YES/NO
ALERTS
ALERT Messages to be displayed
Update (Constant) PRESENCE INFORMATION ON WEBPAGE
PRESENCE UPDATES to be Displayed

THE EVIDENCE

The portal answers four questions at a glance

1. Where is the presence detected now
2. When did the latest event arrive
3. What pattern has build up today
4. Are there any anomalies in pattern

INTELLIGENT ANALYTICS

Bathroom & Bedroom Safety Rule Engine

Duration Tracking

Records exact entry and exit timestamps. Flags stay times exceeding custom safety thresholds.

Frequency Analysis

Monitors day vs. night visit frequency to identify health indicators like sleep disruption or infection.

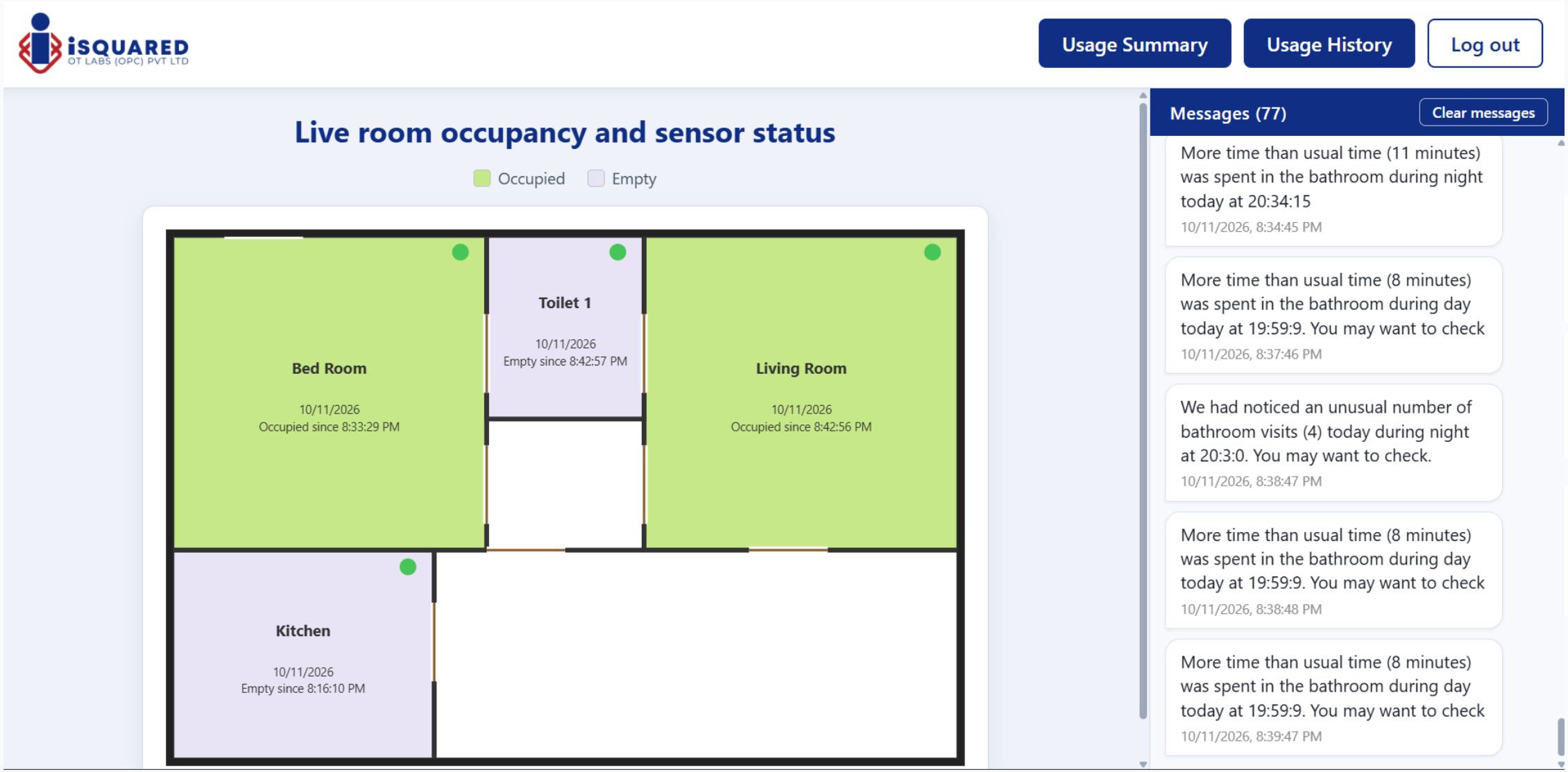
Automated Alerts

Triggers instant messages or dashboard alerts when abnormal behavior is observed

LIVE DEMO

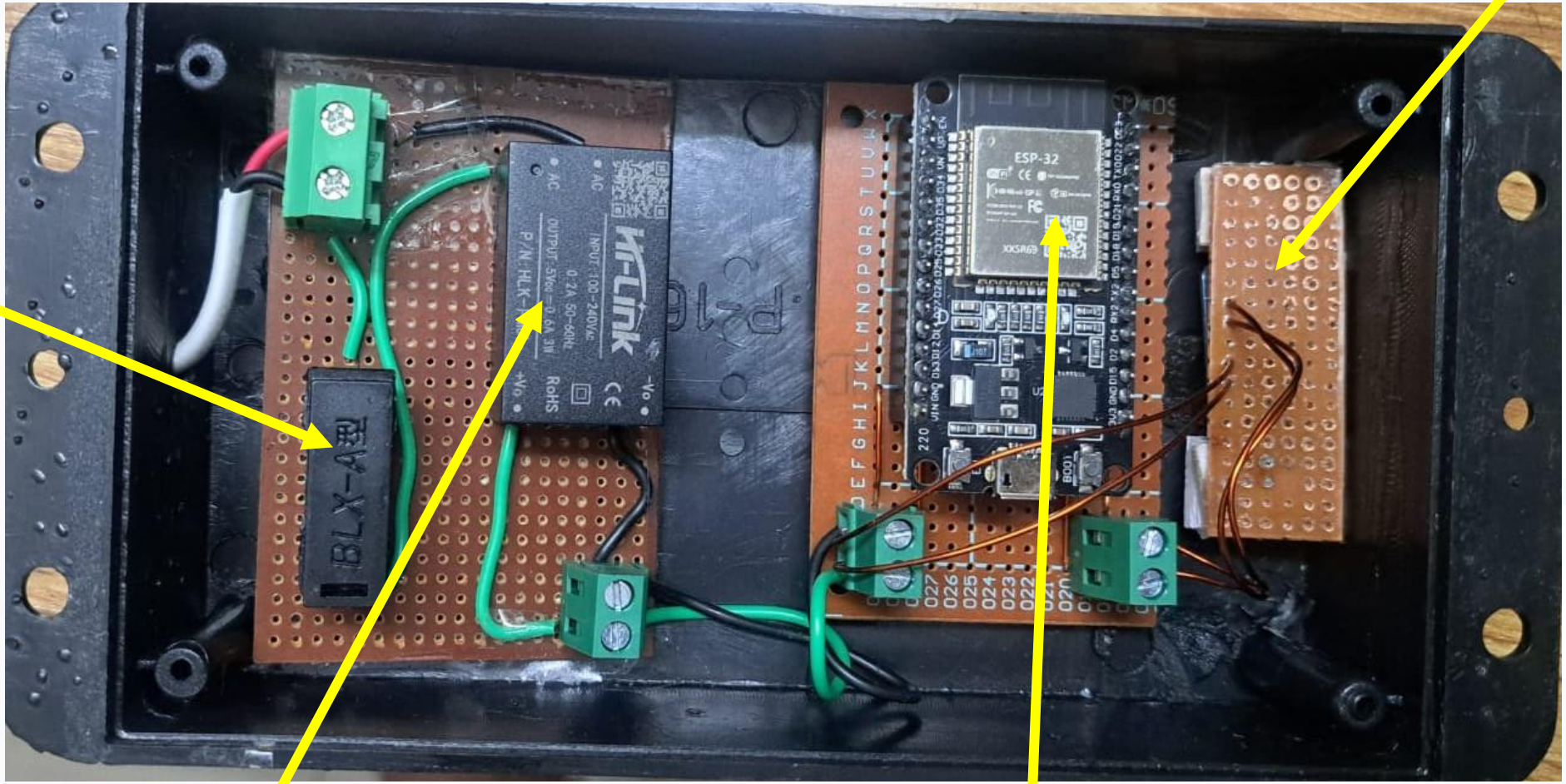
CENTRAL DASHBOARD

Real-Time Activity Verification & Timestamps



SYSTEM PROTOTYPE

Hardware Implementation & Real Demonstration





NEXT STEPS

Future Development Roadmap

Phase 1

Machine Learning Routine Profiling

Automatically baseline individual routine patterns without manual threshold setting.

Phase 2

Multi-Target & Fall Radar Detection

Upgrade to multi-point radar to distinguish multiple occupants and detect sudden height drops.

Phase 3

Caregiver Mobile Application

Native iOS/Android companion app with push notifications and emergency escalation.

Phase 4

Clinical EHR Integration

Export mobility trends to healthcare providers for early detection of cognitive or physical decline.

EMPOWERING INDEPENDENT SENIOR LIVING

Dignity, Safety, and Total Peace of Mind

Gentle Steps proves that effective elder care technology can protect lives without invading personal privacy.

Thank You for Your Consideration